In electronics, the *mean* is commonly called the **DC** (direct current) value. Likewise, **AC** (alternating current) refers to how the signal fluctuates around the mean value. If the signal is a simple repetitive waveform, such as a sine or square wave, its excursions can be described by its peak-to-peak amplitude. Unfortunately, most acquired signals do not show a well defined peak-to-peak value, but have a random nature, such as the signals in Fig. 2-1. A more generalized method must be used in these cases, called the **standard deviation**, denoted by σ (a lower case Greek *sigma*).

As a starting point, the expression, $|x_i - \mu|$, describes how far the i^{th} sample *deviates* (differs) from the mean. The *average deviation* of a signal is found by summing the deviations of all the individual samples, and then dividing by the number of samples, *N*. Notice that we take the absolute value of each deviation before the summation; otherwise the positive and negative terms would average to zero. The average deviation provides a single number representing the typical distance that the samples are from the mean. While convenient and straightforward, the average deviation is almost never used in statistics. This is because it doesn't fit well with the physics of how signals operate. In most cases, the important parameter is not the *deviation* from the mean, but the *power* represented by the deviation from the mean. For example, when random noise signals combine in an electronic circuit, the resultant noise is equal to the combined *power* of the individual signals, not their combined *amplitude*.

The *standard deviation* is similar to the *average deviation*, except the averaging is done with power instead of amplitude. This is achieved by squaring each of the deviations before taking the average (remember, power $\propto$ voltage^2). To finish, the *square root* is taken to compensate for the initial squaring. In equation form, the standard deviation is calculated:

EQUATION 2-2
Calculation of the standard deviation of a signal. The signal is stored in x_i, μ is the mean found from Eq. 2-1, *N* is the number of samples, and s is the standard deviation.

$$\sigma^2 = \frac{1}{N-1}\sum_{i=0}^{N-1}(x_i - \mu)^2$$

In the alternative notation: $\sigma = \sqrt{(x_0 - \mu)^2 + (x_1 - \mu)^2 + \cdots + (x_{N-1} - \mu)^2 / (N-1)}$. Notice that the average is carried out by dividing by $N - 1$ instead of *N*. This is a subtle feature of the equation that will be discussed in the next section. The term, σ^2, occurs frequently in statistics and is given the name **variance.** The standard deviation is a measure of how far the signal fluctuates from the mean. The variance represents the power of this fluctuation. Another term you should become familiar with is the **rms (root-mean-square)** value, frequently used in electronics. By definition, the standard deviation only measures the AC portion of a signal, while the rms value measures both the AC and DC components. If a signal has no DC component, its rms value is identical to its standard deviation. Figure 2-2 shows the relationship between the standard deviation and the peak-to-peak value of several common waveforms.